Ex No: 1b

IMPLEMENTATION OF DEPTH FIRST SEARCH

Aim:

To implement depth first search

Procedure:

Program:

```
# DFS Algorithm Implementation
class Graph:
    def __init__(self, vertices):
        self.vertices = vertices
        self.graph = {i: [] for i in range(vertices)}

    # Function to add an edge to the graph
    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.graph[v].append(u)

    # Recursive function for DFS traversal
    def dfs(self, v, visited):
        print(v, end=" ")  # Print the current node
        visited[v] = True  # Mark the node as visited
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        # Recur for all the adjacent vertices
        for neighbor in self.graph[v]:
            if not visited[neighbor]:
                self.dfs(neighbor, visited)

# Main function to demonstrate DFS
def depth_first_search():
    # Creating a graph with 6 vertices
    g = Graph(6)
        # Adding edges
    g.add_edge(0, 1)
    g.add_edge(0, 2)
    g.add_edge(1, 3)
    g.add_edge(1, 4)
    g.add_edge(2, 5)
    # To track visited nodes
    visited = [False] * g.vertices

    # Start DFS traversal from vertex 0
    print("Depth First Search starting from vertex 0:")
    g.dfs(0, visited)

# Calling the main function
depth_first_search()
```

Output:

```
Depth First Search starting from vertex 0:
0 1 3 4 2 5
```

```
========== RESTART: C:/Users/ASUS/App
pth First Search starting from vertex 0:
1 3 4 2 5
```